

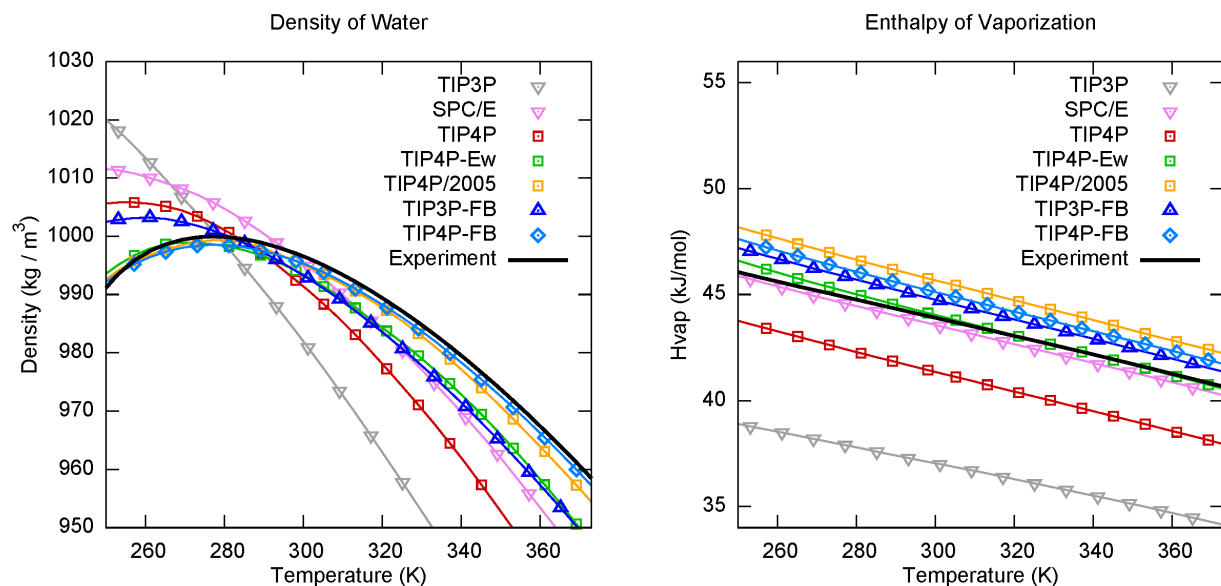
Supporting Information for:
Building Force Fields - an Automatic, Systematic and Reproducible Approach

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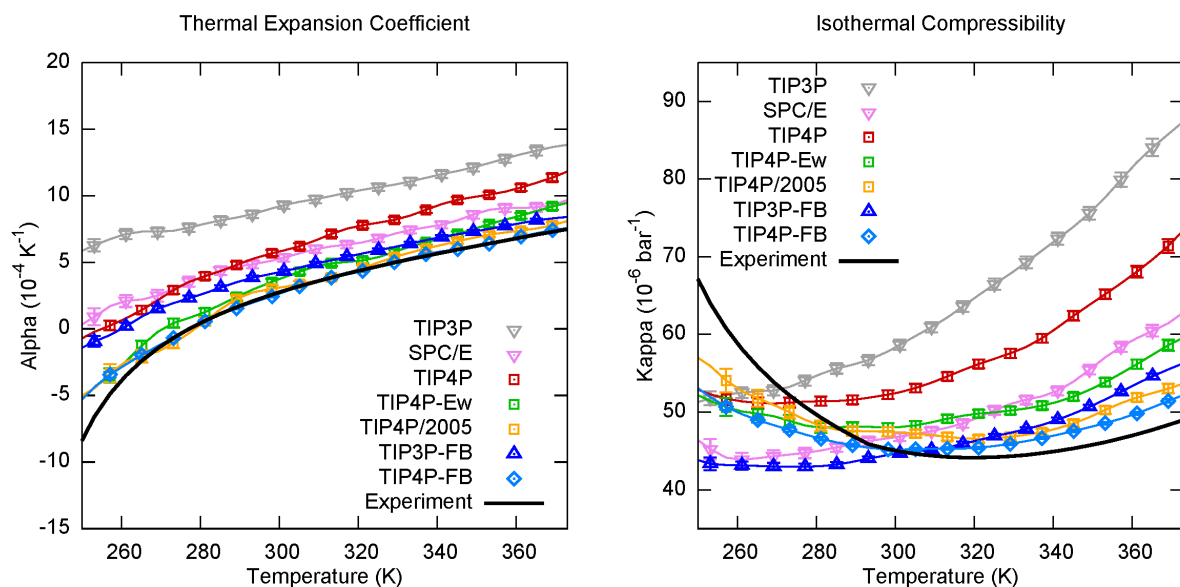
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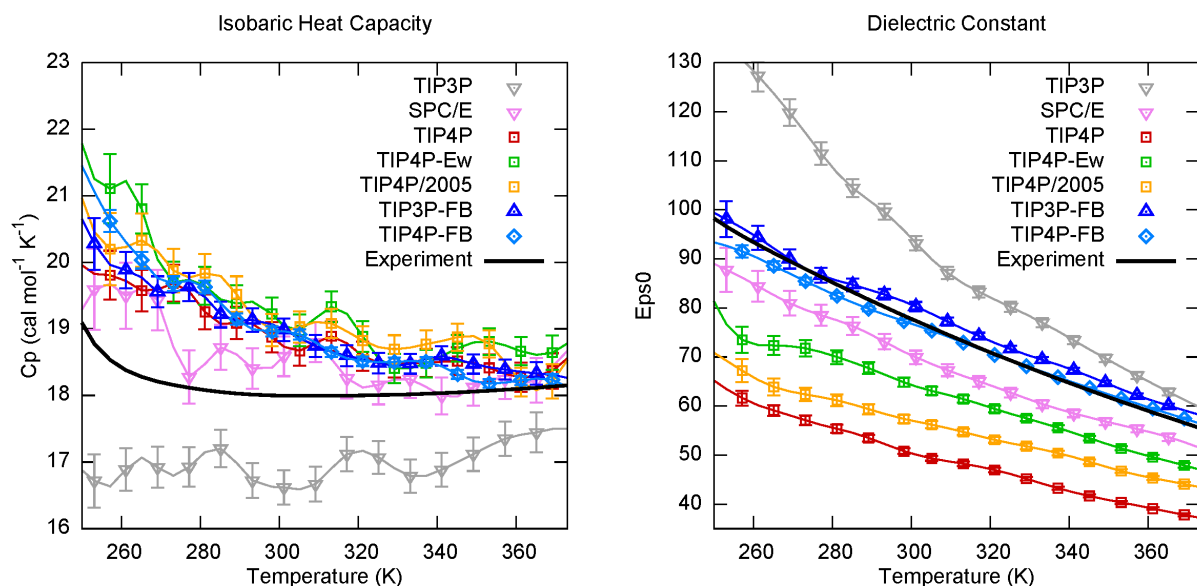
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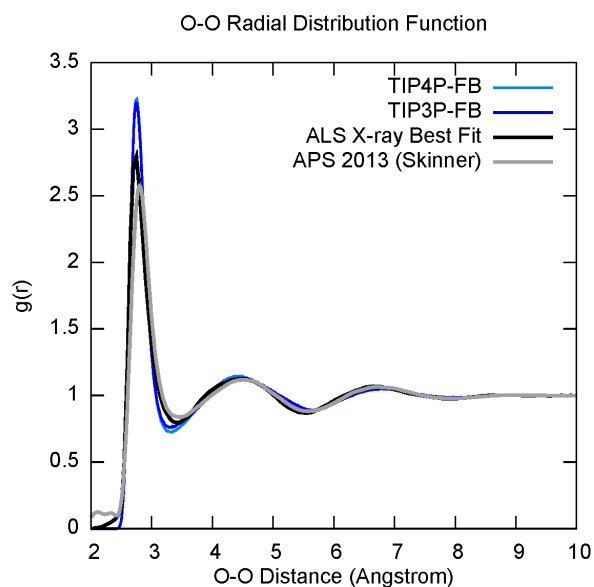
Supporting Figure 1-2. Density and enthalpy of vaporization of liquid water as a function of temperature. Several models from the literature are compared to the experiment, with three-point models shown as triangles and four-point models shown as squares (TIP3P, gray; SPC/E, violet; TIP4P, red; TIP4P-Ew, green; TIP4P/2005, orange). The models presented in this work are TIP3P-FB (dark blue triangles) and TIP4P-FB (light blue diamonds). The simulation uncertainty is smaller than the symbol size in this figure.



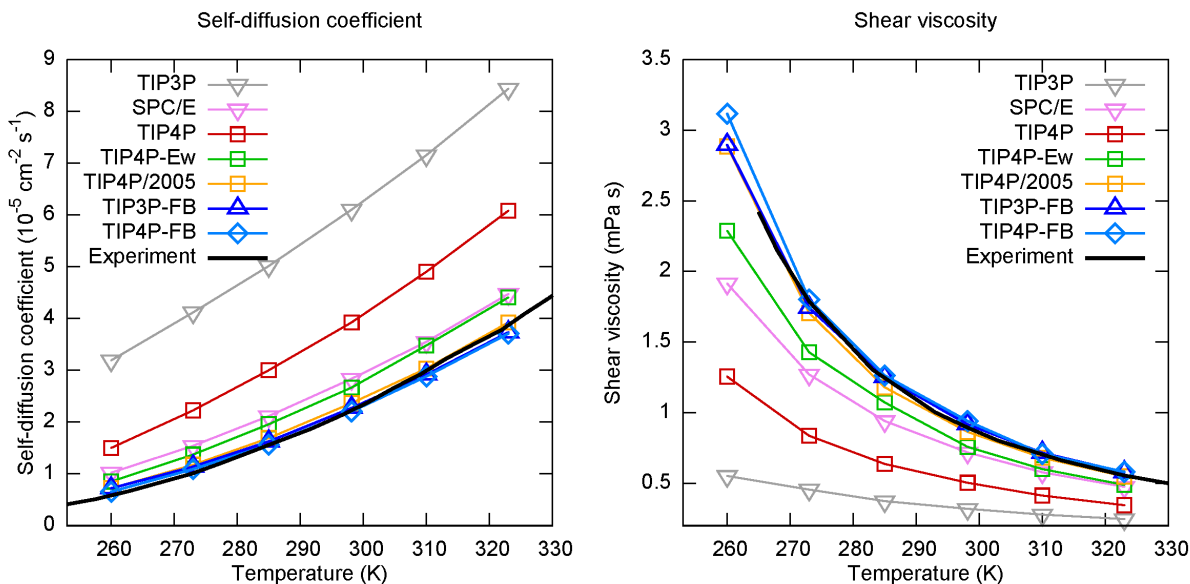
Supporting Figure 3-4. Thermal expansion coefficient and isothermal compressibility of liquid water as a function of temperature. Several models from the literature are compared to the experiment as in Supporting Figure 1. The models presented in this work are TIP3P-FB (dark blue triangles) and TIP4P-FB (light blue diamonds). Error bars represent one standard error.



Supporting Figure 5-6. Isobaric heat capacity and dielectric constant of liquid water as a function of temperature. Several models from the literature are compared to the experiment as in Supporting Figure 1. The models presented in this work are TIP3P-FB (dark blue triangles) and TIP4P-FB (light blue diamonds). Error bars represent one standard error.



Supporting Figure 7. O-O radial distribution function of liquid water at 298.15 K, 1.0 atm. The TIP3P-FB (dark blue) and TIP4P-FB (light blue) models are compared to two experimental determinations (black, gray). The TIP3P-FB and TIP4P-FB models give nearly identical radial distribution functions as indicated by the near-complete overlap of the light blue and dark blue curves.



Supporting Figure 8-9. Self-diffusion coefficient and shear viscosity of liquid water as a function of temperature. Several models from the literature are compared to the experiment as in Supporting Figure 1. The models presented in this work are TIP3P-FB (dark blue triangles) and TIP4P-FB (light blue diamonds). In contrast to Supporting Figure 1-6, these are kinetic properties that were not used in the parameter optimization.

Parameter Name	Units	TIP4P-FB	TIP3P-FB	Prior Width
O-H Bond Length	Å	0.9572	1.0118	0.1
H-O-H Angle	Degree	104.52	108.15	0.1 rad
Oxygen vdW Sigma	Å	3.1655	3.1780	0.1
Oxygen vdW Epsilon	kJ/mol	0.74928	0.65214	0.1
Hydrogen Charge	e	0.52587	0.42422	0.1
Oxygen-VSite Displacement	Å	0.10527	N/A	0.1

Supporting Table 1. Optimized parameters of the TIP4P-FB and TIP3P-FB water models. The TIP4P-FB model geometry is not modified from the initial geometry (cells grayed out), whereas the TIP3P-FB geometry parameters are optimized. The “prior width” indicates the expected variation of the parameter during the optimization and corresponds to the strength of the harmonic penalty function.

Temp. (K)	Pressure	Density (kg m ⁻³)	ΔH_{vap} (kJ mol ⁻¹)	Quantum ΔH_{vap} Correction	Nonideality ΔH_{vap} Correction	Thermal Expan. (10 ⁻⁴ K ⁻¹)	Isothermal Compress. (10 ⁻⁶ bar ⁻¹)	Isobaric Heat Capacity (cal/mol/K)	Quantum Heat Capacity Correction	Dielectric Constant
249.15	1.0 atm	990.497	46.103	-0.003	-0.003	-8.877	67.985	19.177	-2.720	98.650
253.15	1.0 atm	993.547	45.919	-0.004	-0.004	-6.606	63.996	18.777	-2.673	96.580
257.15	1.0 atm	995.816	45.740	-0.005	-0.005	-4.874	60.783	18.532	-2.627	94.683
261.15	1.0 atm	997.476	45.565	-0.005	-0.005	-3.506	58.116	18.380	-2.581	92.898
265.15	1.0 atm	998.647	45.393	-0.006	-0.006	-2.395	55.851	18.280	-2.536	91.186
269.15	1.0 atm	999.414	45.223	-0.008	-0.008	-1.469	53.893	18.210	-2.492	89.525
273.15	1.0 atm	999.840	45.053	-0.009	-0.009	-0.680	52.176	18.157	-2.447	87.903
277.15	1.0 atm	999.972	44.884	-0.011	-0.011	0.003	50.653	18.115	-2.405	86.311
281.15	1.0 atm	999.848	44.714	-0.013	-0.013	0.604	49.288	18.080	-2.362	84.749
285.15	1.0 atm	999.497	44.544	-0.015	-0.015	1.141	48.056	18.052	-2.319	83.213
289.15	1.0 atm	998.943	44.374	-0.018	-0.018	1.626	46.934	18.030	-2.278	81.705
293.15	1.0 atm	998.204	44.203	-0.021	-0.021	2.068	45.892	18.015	-2.237	80.223
298.15	1.0 atm	997.045	43.989	-0.025	-0.025	2.572	45.247	18.002	-2.186	78.409
301.15	1.0 atm	996.234	43.861	-0.028	-0.028	2.853	44.943	17.998	-2.155	77.339
305.15	1.0 atm	995.026	43.689	-0.032	-0.032	3.206	44.622	17.995	-2.117	75.935
309.15	1.0 atm	993.684	43.517	-0.037	-0.037	3.539	44.390	17.995	-2.078	74.556
313.15	1.0 atm	992.216	43.345	-0.043	-0.043	3.853	44.239	17.996	-2.039	73.200
317.15	1.0 atm	990.628	43.172	-0.049	-0.049	4.152	44.162	17.999	-2.000	71.869
321.15	1.0 atm	988.927	42.999	-0.056	-0.056	4.438	44.153	18.002	-1.963	70.560
325.15	1.0 atm	987.119	42.826	-0.064	-0.064	4.712	44.209	18.006	-1.926	69.275
329.15	1.0 atm	985.208	42.652	-0.072	-0.072	4.975	44.324	18.011	-1.889	68.012
333.15	1.0 atm	983.199	42.476	-0.081	-0.081	5.231	44.496	18.018	-1.853	66.772
337.15	1.0 atm	981.095	42.300	-0.091	-0.091	5.478	44.723	18.026	-1.817	65.555
341.15	1.0 atm	978.900	42.123	-0.103	-0.103	5.719	45.003	18.035	-1.781	64.359
345.15	1.0 atm	976.617	41.943	-0.115	-0.115	5.954	45.333	18.046	-1.747	63.184
349.15	1.0 atm	974.249	41.762	-0.128	-0.128	6.185	45.714	18.058	-1.712	62.030
353.15	1.0 atm	971.798	41.580	-0.142	-0.142	6.411	46.143	18.071	-1.677	60.897
357.15	1.0 atm	969.266	41.396	-0.158	-0.158	6.633	46.621	18.086	-1.644	59.784
361.15	1.0 atm	966.655	41.210	-0.175	-0.175	6.853	47.148	18.101	-1.611	58.691
365.15	1.0 atm	963.966	41.024	-0.194	-0.194	7.071	47.722	18.117	-1.578	57.617
369.15	1.0 atm	961.202	40.837	-0.214	-0.214	7.287	48.346	18.134	-1.545	56.561
373.15	1.0 atm	958.364	40.652	-0.235	-0.235	7.501	49.019	18.151	-1.512	55.522
380.00	1.3 atm	953.327	40.322	0.384	-0.277	7.876	50.205	18.195	-1.457	53.796
390.00	1.8 atm	945.624	39.830	0.451	-0.347	8.410	52.326	18.261	-1.379	51.363
400.00	2.4 atm	937.486	39.323	0.516	-0.430	8.951	54.786	18.337	-1.303	49.037
410.00	3.3 atm	928.921	38.798	0.578	-0.527	9.505	57.641	18.425	-1.230	46.812
420.00	4.3 atm	919.929	38.254	0.638	-0.641	10.078	60.905	18.525	-1.157	44.682
430.00	5.6 atm	910.507	37.688	0.697	-0.773	10.679	64.630	18.639	-1.087	42.642
440.00	7.2 atm	900.649	37.099	0.753	-0.924	11.314	68.948	18.769	-1.017	40.687
450.00	9.2 atm	890.341	36.483	0.807	-1.096	11.992	73.844	18.918	-0.950	38.810
298.15	1 kbar	1037.872				3.502	35.723	17.123	-2.186	81.900
298.15	2 kbar	1071.908				4.044	29.215	16.703	-2.186	85.015
298.15	3 kbar	1101.035				4.305	24.666	16.492	-2.186	87.870
298.15	4 kbar	1126.580				4.407	21.371	16.375	-2.186	90.545
298.15	5 kbar	1149.422				4.425	18.888	16.301	-2.186	93.087
298.15	6 kbar	1170.147				4.400	16.931	16.251	-2.186	95.529
298.15	7 kbar	1189.159				4.355	15.355	16.216	-2.186	97.891
298.15	8 kbar	1206.748				4.300	14.050	16.195	-2.186	100.184
298.15	9 kbar	1223.131				4.241	12.949	16.186	-2.186	102.416

Supporting Table 2. Properties of water used to optimize TIP4P-FB and TIP3P-FB. The temperatures are linearly spaced with a spacing of 4 K up to the boiling point, and 10 K beyond the boiling point up to 450 K. High-pressure simulations are linearly spaced. The nonideality and quantum corrections are generated following procedures from Ref. 1.

Ice Phase	Temp. (K)	Pressure (bar)	Density (kg m ⁻³)
Ih	250.0	0	920.0
II	225.0	2800	1195.7
III	250.0	2800	1165.0
V	223.0	5300	1283.0
VI	225.0	11000	1373.0

Supporting Table 3. Experimental densities of ice phases used to optimize TIP4P-FB. This data was left out of the TIP3P-FB parameterization due to the inability of 3-point models to accurately describe the properties of the high-pressure ice phases.

Computational Details

In each nonlinear iteration of the optimization procedure, ForceBalance evaluates the objective function contribution from each *target* (a reference data set paired with a simulation protocol). For each target, ForceBalance also calculates the derivative of the objective function with respect to the force field parameters. Most of these derivatives are functions of derivatives of the potential energy, which are evaluated numerically using a central finite difference approximation. The Gauss-Newton approximation is used to form an approximate objective function Hessian from the first derivatives of the simulated properties. The targets executed by ForceBalance are:

1) *Condensed phase properties.* The experimental reference data is taken from the IAPWS-95 formulation of pure water properties.² In order to simulate the properties, ForceBalance launches isothermal-isobaric (NPT) simulations of liquid water and ice at temperatures and pressures specified in Supporting Tables 2 and 3 (also at <https://simtk.org/home/tip3p-fb>). The simulations use the OpenMM software³ running on the CUDA platform. The Work Queue library allows ForceBalance to act as a distributed computing server, and the simulations are executed remotely on GPU clusters.

The NPT simulations for optimizing condensed phase properties contain 216 water molecules. The simulation length is 4 ns with a 2.0 fs time step (2000000 steps) using a leapfrog integrator, and coordinates are saved every 1 ps (5000 steps). We use Langevin dynamics with a time constant of 1.0 ps⁻¹ and a Monte Carlo barostat with a volume adjustment frequency of 10 fs (20 steps). The van der Waals interactions are truncated using a 9.0 Angstrom cutoff, and an isotropic long-range correction is applied. Long-range electrostatics are treated using particle mesh Ewald (PME) with a real space cutoff of 7.0 Angstrom. After the simulation is finished running, the trajectories are post-processed to calculate the condensed phase properties and their derivatives. This information is transmitted back to the ForceBalance program which evaluates the objective function / gradient for condensed phase properties and moves onto the next target.

2) *Ab initio potentials and forces.* The cluster geometries for the reference energy and force calculations were periodically extracted from simulations of iAMOEBA liquid-gas interfaces and liquid-solid interfaces. 7,200 clusters were extracted for each cluster size ranging from 2 – 22 following this procedure. The time intervals were offset for each cluster size to minimize the possibility of sampling the same water molecules twice. The water molecules in the clusters

were reset to the TIP3P geometry ($d_{\text{OH}} = 0.9572$, $\theta_{\text{HOH}} = 104.52^\circ$) prior to the energy and force calculations.

The liquid-gas interface simulations were performed in the NVT ensemble and used a modified Beeman integrator with the Bussi thermostat for temperature control. We prepared a $30 \times 30 \times 100$ Å rectangular box and performed simulations from 250 – 640 K in 10 K increments. The length of each simulation was 5 ns. Clusters were sampled from the simulation by choosing the closest several water molecules to a random point in the simulation cell. The sampling scheme produces diverse clusters with bulk liquid, surface, and supercritical character.

The liquid-solid coexistence simulations begins from a simulation cell containing half water and half ice. The size of the simulation cell depended on the particular phase of ice being studied, ranging from 864 molecules (ice Ih) to 1280 molecules (ice VI). With the exception of ice II, all of the ices in this study are proton-disordered, so the simulation cell cannot be directly taken from crystallographic data. We implemented the algorithm of Buch et al.⁴ which generates proton-disordered configurations and performs annealing until the Bernal-Fowler rules were satisfied. For ice III and V, special care was taken to ensure the statistically correct occupations of the nonequivalent hydrogen sites.⁵

A crucial component of the melting point calculation is a barostat that supports anisotropic box deformations, because the initial condition has a different aspect ratio than a fully frozen configuration. In particular, ice V has a monoclinic unit cell, which requires a barostat capable of deforming the angles of the simulation cell in addition to the lattice lengths. For this purpose, we implemented an anisotropic Monte Carlo barostat into TINKER. These simulations used the modified Beeman integrator and the Bussi thermostat in addition to the anisotropic Monte Carlo barostat.

The phase points chosen for sampling corresponded to trajectories that still had a phase boundary after 1 ns of simulation time, so they are near the liquid-solid phase boundary. The rationale was to sample some geometries from the solid, some from the liquid (at low temperature / high pressure), and some from the interface. The exception was ice VII, which did not have any stable interfaces, so snapshots were taken from a simulation with mild position restraints.

For each cluster size, we extracted 7,200 clusters from the following simulations:

Liquid/Gas: 3600 total	=	90 snapshots x 40 phase points (250-640 K every 10 K, NVT)
Liquid/Ice-Ih: 600 total	=	100 snapshots x 6 phase points (250-260 K every 2 K, 1.0 atm)
Liquid/Ice-II: 600 total	=	50 snapshots x 12 phase points (215-225K every 2 K, 2500 bar; 220-230K every 2 K, 5300 bar)
Liquid/Ice-III: 600 total	=	20 snapshots x 30 phase points (224-238K every 2 K, 2500 bar; 224-244K every 2 K, 5300 bar;

Liquid/Ice-V: 600 total		=	224-244K every 2 K, 10856 bar) 20 snapshots x 30 phase points (220-236K every 2 K, 5300 bar; 233-237K every 1 K, 5300 bar; 257-260K every 1 K, 10856 bar, 240-262K every 2 K, 10856 bar)
Ice-VI:	600 total	=	300 snapshots x 2 temperatures (260K, 261K, 10856 bar)
Ice-VII:	600 total	=	600 snapshots x 1 phase point (350K, 100000 bar)

Energies and gradients were calculated for these clusters at the dual basis RI-MP2/heavy-aug-cc-pVTZ level of theory as discussed in the Methods section. The full ab initio data set is provided online (<https://simtk.org/home/tip3p-fb>). An example Q-Chem input file for one such calculation is given below.

During the optimization, ForceBalance calculates the potential energy, force, and parameteric derivatives for all geometries in the cluster using the force field; these quantities are used to calculate the contribution to the objective function and its derivatives.

Parameter Optimization: Once ForceBalance finishes evaluating the contributions to the objective function from each target, it takes an optimization step using the Levenberg-Marquardt (trust-radius Newton-Raphson) algorithm.⁶⁻⁷ The trust radius is set to 0.2 in the rescaled space of optimization parameters. Due to statistical noise in the condensed phase calculation, the objective function for a given optimization step may increase. When this occurs, the optimization step is rejected and new simulations are run using the previous parameter values. The simulation data is appended to the existing data at these values of the parameters (from the previous optimization cycle) and used to improve the optimization step. The simulations for subsequent optimization cycles are doubled in length; thus, the simulation length is adaptively increased as the optimization nears convergence. Each nonlinear iteration takes 2 to 12 hours depending on the simulation length, and the objective function is converged to within a standard error of 1-3 after ~30 optimization cycles.

\$comments
Sample Q-Chem energy and gradient calculation for: Water cluster DB-RIMP2/HATZ
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O 1.0422652391 1.5560120579 -0.8270515493
H 1.3790694980 2.4519391196 -0.8375124950
H 1.8150822451 1.0075157889 -0.9617273013
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Calculation of O-O radial distribution function and kinetic properties. The radial distribution function and kinetic properties are not part of the optimization data set and serve as validation for our derived parameters. The validation calculations are performed in GROMACS.⁸ We first ran a number of simulations in the NPT ensemble for system sizes consisting of 216, 343, 512, 1,000, 2,500, 4,000, and 16,000 molecules; these simulations used a leapfrog algorithm to integrate Newton's equations of motion with a 2.0 fs time step. The SETTLE algorithm was applied to keep the molecular geometry constant. The electrostatic interactions were treated using the particle-mesh Ewald algorithm with a real-space cutoff of 9.0 Angstrom, and the Lennard-Jones interactions were treated using a potential-switching scheme with the forces going smoothly to zero at a 9.0 Angstrom cutoff, with an isotropic long-range correction added. The Bussi thermostat with a time constant of $\tau=10$ ps was used to provide temperature control without unduly altering the kinetics, and the Parrinello-Rahman barostat with a time constant of $\tau=10$ ps was used to provide pressure control. A total of 100 ns of time evolution was followed for each system size and the temperatures 260.00, 273.00, 285.00, 298.15, 310.00 and 323.00 K at atmospheric pressure; positions and velocities were periodically sampled at 10 ps intervals. The O-O radial distribution function was calculated using samples from the 298.15 K, 16,000 molecule simulation.

The diffusion coefficient was calculated by running a large number of energy-conserving (NVE) simulations using evenly spaced NPT-sampled coordinates and velocities as initial conditions, then applying the Einstein relation which relates the diffusion constant to the time dependence of the mean squared displacement. We performed 50 to 1,000 NVE simulations per NPT simulation depending on the box size (50 simulations per temperature for the 16,000 molecule system to 1,000 simulations per temperature for the 216 molecule system). The NVE simulations were performed using a double precision build of GROMACS; we used the same simulation settings as for the NPT simulations with some modifications: (1) the time step was decreased to 1.0 fs, (2) the temperature and pressure control algorithms were removed, and (3) the neighbor list was updated at every time step. This was sufficient to keep energy drift down to ~ 0.1 kJ/mol (0.01% of the NPT energy fluctuation) for the largest system. At each temperature, the diffusion constant was extrapolated to the infinite-size limit⁹ and used as the final result. We calculated the shear viscosity in two different ways – from the slope of the diffusion coefficient extrapolation, and from the off-diagonal components of the NPT pressure tensor using the Green-Kubo relation.⁹ Both methods of calculating the shear viscosity gave highly similar results, though the Green-Kubo relation gave much smaller error bars.

References

(1) Horn, H. W.; Swope, W. C.; Pitner, J. W.; Madura, J. D.; Dick, T. J.; Hura, G. L.; Head-

Gordon, T. Development of an Improved Four-Site Water Model for Biomolecular

Simulations: Tip4p-Ew. *J. Chem. Phys.* **2004**, *120*, 9665-9678.

- (2) Wagner, W.; Pruss, A. The Iapws Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use. *J. Phys. Chem. Ref. Data* **2002**, *31*, 387-535.
- (3) Eastman, P.; Friedrichs, M. S.; Chodera, J. D.; Radmer, R. J.; Bruns, C. M.; Ku, J. P.; Beauchamp, K. A.; Lane, T. J.; Wang, L.-P.; Shukla, D., et al. Openmm 4: A Reusable, Extensible, Hardware Independent Library for High Performance Molecular Simulation. *J. Chem. Theory Comput.* **2013**, *9*, 461-469.
- (4) Buch, V.; Sandler, P.; Sadlej, J. Simulations of H₂O Solid, Liquid, and Clusters, with an Emphasis on Ferroelectric Ordering Transition in Hexagonal Ice. *J. Phys. Chem. B* **1998**, *102*, 8641-8653.
- (5) MacDowell, L. G.; Sanz, E.; Vega, C.; Abascal, J. L. F. Combinatorial Entropy and Phase Diagram of Partially Ordered Ice Phases. *J. Chem. Phys.* **2004**, *121*, 10145-10158.
- (6) Levenberg, K. A Method for the Solution of Certain Non-Linear Problems in Least Squares. *Q. Appl. Math.* **1944**, *2*, 164-168.
- (7) Marquardt, D. W. An Algorithm for Least-Squares Estimation of Nonlinear Parameters. *J. Soc. Ind. Appl. Math.* **1963**, *11*, 431-441.
- (8) Pronk, S.; Pall, S.; Schulz, R.; Larsson, P.; Bjelkmar, P.; Apostolov, R.; Shirts, M. R.; Smith, J. C.; Kasson, P. M.; van der Spoel, D., et al. Gromacs 4.5: A High-Throughput and Highly Parallel Open Source Molecular Simulation Toolkit. *Bioinformatics* **2013**, *29*, 845-854.
- (9) Yeh, I. C.; Hummer, G. System-Size Dependence of Diffusion Coefficients and Viscosities from Molecular Dynamics Simulations with Periodic Boundary Conditions. *J. Phys. Chem. B* **2004**, *108*, 15873-15879.

